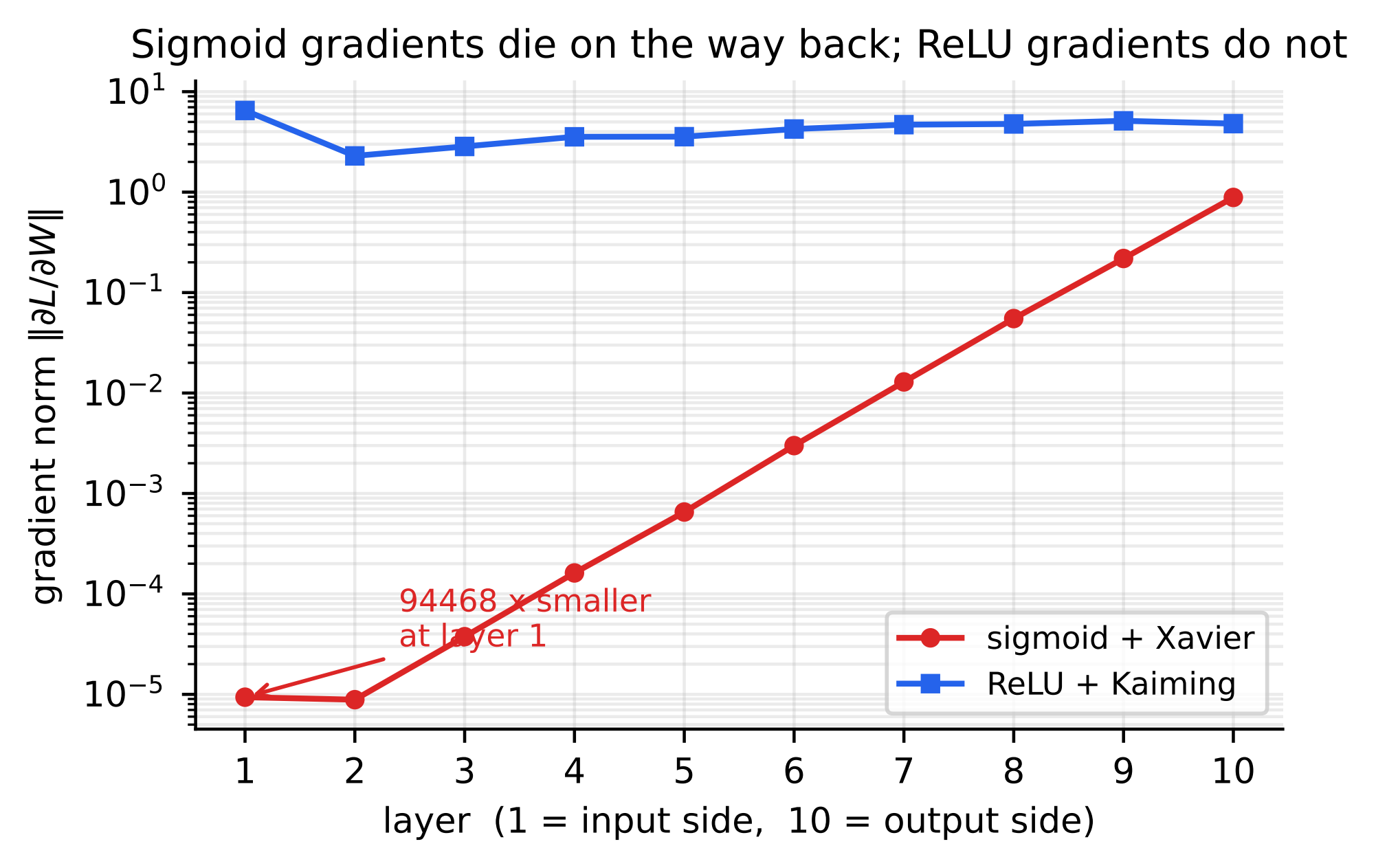
Sigmoid gradients die on the way back; ReLU gradients do not
gradient norm ‖∂L/∂W‖
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
94468 x smaller
at layer 1
sigmoid + Xavier
ReLU + Kaiming
1
2
3
4
5
6
7
8
9
10
layer (1 = input side, 10 = output side)